Decoding Citi Bike Demand: The Impact of Weather, Seasonality and Holidays in NYC

Advanced Data Science · Project Pre-Submission

1. PROJECT AND GROUP

Group 14: Harshit Kumar (240470) · Ishaan Jatain (240484) · Himanshu Yadav (_____) · **Code:** github.com/harshitkumar24cse/citibike-demand

2. STAKEHOLDER AND PROBLEM

Stakeholder: Citi Bike operations and rebalancing team (NYC). **Problem:** demand swings with weather, season and holidays, but bikes are rebalanced on fixed routines, so stations run empty or full. **Why it matters:** this causes lost trips and wasted truck runs; knowing how demand shifts lets the team move bikes ahead of time.

3. RESEARCH QUESTIONS AND HYPOTHESES

RQ1 How much do weather and the calendar change hourly demand? **RQ2** Do members and casual riders respond differently?

- **H1** Warmer hours have more trips. **H2** Rainy hours have fewer trips than dry hours at similar temperatures.
- **H3** Casual riders are more weather-sensitive than members. **H4** Casual trips are longer than member trips.
- **H5** Weekends have weaker commute peaks and a higher casual share than weekdays.

4. DATA SOURCES

Source and link	Coverage and rows	Key variables
Citi Bike trips · citibikenyc.com/system-data	Apr, Jun, Oct, Dec 2025; 15.3 M trips, 25% sample	Times, stations, coordinates, bike type, rider type
Open-Meteo weather · open-meteo.com	Same months, 2,928 hours	Temperature, rain, wind, humidity, pressure, cloud
NOAA daily records · ncei.noaa.gov	Same months, 122 days	Max/min temperature, rain (validation only)
Citi Bike GBFS stations · gbfs.citibikenyc.com	~2,500 stations (current)	Capacity, location
US/NY holidays · Python holidays	14 in 2025 (3 in study months)	Holiday date and name

Rows: 3,822,094 trips × 41 variables. **Constraints:** ~3 GB of trip files (seeded 25% sample for 8 GB RAM); no NOAA hourly data for Dec 2025, so Open-Meteo is used throughout and checked against NOAA (temperature r = 0.99, wet days agree 92%); station data is a current snapshot (96.6% match).

5. INTEGRATION PLAN (SSOT)

- **Keys:** one row per trip; start hour → weather; station ID → capacity; date → holiday. Weather converted from UTC to New York time.
- **Cleaning:** duplicate ride IDs dropped (0); trips under 1 or over 180 min (4,410) and outside the study months (361) removed.
- **Missing values:** weather gaps up to 3 h interpolated (none); unmatched stations and coordinates left empty.
- **Units:** °C, mm, m/s, hPa, %, km, min, km/h. **Checks:** row count unchanged after joins; unmatched weather 0%, stations 3.4%.

6. METHODS AND BADIR WORKFLOW

Business question	When, and by how much, do weather and the calendar shift demand?
Analysis plan	Trips per hour; EDA and correlation; t-test, Mann-Whitney U, Kruskal-Wallis, chi-square; multiple regression on weather and calendar.
Data collection	build_ssot.py pipeline builds the SSOT (done).
Insights	Effect sizes with confidence intervals; hypotheses rated supported, not supported or inconclusive.
Recommendations	Rebalancing and staffing rules for the operations team.

7. VALIDATION AND DECISION VALUE

Each hypothesis gets a test with effect size, a chart and a robustness check (per month, weekdays only, non-parametric); unsupported results are valid outcomes. **Decision value:** change in trips per °C, rainy hour and day type → extra rebalancing before warm, dry weekends; lighter dispatch in rain or freezing cold; priority for large stations.

8. LIMITATIONS AND NEXT STEPS

- **Assumptions:** one weather point for the city; under 0.1 mm/h counts as dry; 1-180 min trips are genuine.
- **Constraints:** one month per season; trips understate demand at empty or full stations; station snapshot; only 3 holidays (described, not tested); association, not causation.
- **Milestones:** Stage 1 problem, data, SSOT (done) → Stage 2 quality report, EDA, tests → Stage 3 findings, recommendations, roadmap.
- **Deviations:** none; all requirements met (≥100,000 rows, ≥20 variables, multiple sources, no Kaggle).